# Afferent Health

Using the data health systems already collect to save lives and make care more efficient.

CAPABILITY STATEMENT

NEW YORK, NY

## WHO WE ARE

Afferent turns the data a health system already collects into decisions that save lives and reduce cost. Hospitals record far more about a patient than anyone ever reads. We learn from the whole of it - waveforms, monitor streams, labs, and the electronic health record - and return predictions a clinician can act on and defend.

We work where the data are richest and the stakes are highest: surgery and intensive care. Death, delirium, unplanned ICU transfer, kidney injury and respiratory failure are among the most expensive events in medicine, and the most preventable when seen early. Chronic pain and long-term opioid use begin in the same window and are visible in the same record.

We choose the architecture that performs best, including high-capacity models that learn feature combinations no human would specify. Not every such model can be made fully explainable, and we do not pretend otherwise; what we guarantee is that it is accountable. Where a model admits explanation, we can name the clinical and physiologic features it is using; where it does not, we say so and earn trust through physiologic checks, negative controls, and disclosed limits. We integrate practical clinical experience with AI modeling expertise to deliver products that are clinically implementable as well as scientifically sound.

## CORE COMPETENCIES

- **We make the data you already collect more useful.** No new hardware and no new workflow. The signals are already recorded; we turn them into earlier warning, which means fewer patients escalated to a higher level of care, fewer deaths, and lower cost per case.
- **Risk prediction around surgery and in intensive care.** Death, delirium, unplanned ICU transfer, kidney injury, airway and breathing failure, and the lasting pain and long-term opioid use that can follow an operation.
- **Accountable clinical AI.** Negative-control validation, physiologic sanity-checking, and distillation analysis - the transparency and traceability principles of Executive Order 13960.
- **Multimodal waveform and signal processing.** Learning from the shape of the raw waveforms - arterial pressure, exhaled CO2, pulse oximetry, EEG - not just the numbers a monitor summarises them into.
- **Enterprise-scale clinical data.** We supply both halves of the job: the cohort definition, record linkage, and governance that large multi-site data demand, and the modelling that follows. That is what an enterprise warehouse such as VA's Corporate Data Warehouse requires.
- **Regulatory-defensible evidence.** Rigorous anti-leakage and external validation, plus the negative controls, confound disclosure and peer-reviewed reporting that de-risk a model for procurement and deployment.
- **Public-private delivery, built to a measurable return.** We operate comfortably as a commercial partner inside a government program and build to a case the sponsor can measure: the complications we predict are among the costliest in medicine, so earlier warning avoids spend as well as harm.

## DIFFERENTIATORS

- **Physician-scientist expertise to translate models into practice.** An NIA-funded MD-PhD anesthesiologist PI, a senior machine-learning-methods PI, and an MD/PhD-candidate inventor working directly on perioperative signals. Engineering-only teams cannot field this combination, and it is the difference between a model that validates and one that gets used.
- **Peer-reviewed science plus proprietary methods.** Published clinical and computational work underwrites our methods. Our team has built both proprietary and openly published machine-learning methods for physiologic data, and is practiced at explaining them to regulators and reviewers in the terms those audiences require.
- **We measure the problem before selling the solution.** The VA surgical population is 60.0% frail or very frail against 21.3% in the private-sector NSQIP cohort (*JAMA Surgery* 2022), the registry the standard surgical risk calculators are built from. Our first deliverable on any new population is a calibration audit, not a model.

## KEY PERSONNEL

- **D.K. Adrian Williams, MSc** - physician-scientist in training (MD/PhD candidate, Albert Einstein College of Medicine); inventor and modeling lead. Inventor of a calibration-free correction for pulse-oximetry skin-tone bias, and of novel methods in clinical capnography, EEG, and multimodal physiologic signal processing.
- **Joseph R. Scarpa, MD-PhD** - Assistant Professor of Anesthesiology, Weill Cornell Medicine, and Englander Institute for Precision Medicine; NIA-funded principal investigator (R03 AG095722, biological age and surgical outcomes); perioperative machine learning.
- **Ruben Coen-Cagli, PhD** - Associate Professor, Albert Einstein College of Medicine. PhD in physics; 25 years developing machine-learning methods and theory for high-dimensional signals. Continuously NIH-funded principal investigator (two NEI R01s and an NIDA RF1), with NSF support and an NVIDIA early-adopter award.

## PAST PERFORMANCE

> Afferent is a newly formed firm. Past performance is scientific.

- **Development and validation of a multi-modal contactless sensing system for surgical risk analysis in a real-world environment.** Scarpa JR, et al. *PLOS Digital Health*. PMID 41231841 *End-to-end build-and-validate of perioperative risk prediction in a live setting.*
- **Domain-aware interpretable machine learning model for predicting postoperative hospital length of stay from perioperative data: a retrospective observational cohort study.** Hussain I, Scarpa JR, Boyer R. *Bioengineering (Basel)*. PMID 41749687 *Interpretable prediction of a direct cost driver.*
- **Biological aging increases risk of postoperative morbidity and mortality: an international, multi-cohort study.** Scarpa JR, et al. *npj Aging*. PMID 42401606 *Risk stratification that targets prevention where it matters most.*
- **Relating natural image statistics to patterns of response covariability in macaque primary visual cortex.** Farzmahdi A, Kohn A, Coen-Cagli R. *Nature Communications*. PMID 40695825 *The rigor behind extracting reliable signal from noisy, high-dimensional data.*
- **Response sub-additivity and variability quenching in visual cortex.** Goris RLT, Coen-Cagli R, et al. *Nature Reviews Neuroscience*. PMID 38374462
- **Clinical inventions under filing.** Calibration-free correction of pulse-oximeter skin-tone bias. *A safety and equity-corrective monitoring capability the government is actively seeking.*

| BUSINESS | PRIMARY NAICS | ADDITIONAL NAICS |
|---|---|---|
| Clinical AI research, development, and consulting. Small business. New York, NY. | **541512** Computer Systems Design Services *(primary NAICS of the VA T4NG2 IDIQ)* | 541511 Custom Computer Programming Services; 541715 Research and Development in the Physical, Engineering, and Life Sciences (except Nanotechnology and Biotechnology); 541690 Other Scientific and Technical Consulting Services |

*We bring the clinical science and the physician-scientist bench; our integrator partners bring the vehicles, the infrastructure, and the delivery.*